



For an Open Mind

UNIVERSAL TECHNOLOGY AND MANAGEMENT UNIVERSITY

UTAMU

UNIVERSAL TECHNOLOGY AND MANAGEMENT UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE

**AN AI-ENHANCED WEB-BASED PRISON MANAGEMENT SYSTEM FOR
MANAGING INMATE AND VISITOR RECORDS: A CASE STUDY OF LUZIRA
PRISON, UGANDA**

By

MUHANGUZI AMOS

SEPT23/BCS/3584U/F

muhanguzi.amos@student.utamu.ac.ug

&

NAKALEMBE SARAH

SEPT23/BCS/3630U/H

nakalembe.sarah@student.utamu.ac.ug

SUPERVISED BY

MR. ALEX MUKAMA

amukama@utamu.ac.ug

**A PROPOSAL SUBMITTED TO THE SCHOOL OF COMPUTING AND ENGINEERING IN
PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF A BACHELOR'S
DEGREE IN COMPUTER SCIENCE OF UNIVERSAL TECHNOLOGY AND MANAGEMENT
UNIVERSITY**

JULY, 2026

GROUP MEMBERSHIP

S/N	NAME	REGISTRATION NUMBER	SIGNATURE
01	MUHANGUZI AMOS	SEPT23/BCS/3584U/F	
02	NAKALEMBE SARAH	SEPT23/BCS/3630U/H	

DECLARATION

We, Muhanguzi Amos and Nakalembe Sarah, declare that this project proposal is our original work and has not been submitted anywhere for any academic award. All sources and references used have been duly acknowledged and cited.

Signature: 

Date:

MUHANGUZI AMOS

Signature: 

Date:

NAKALEMBE SARAH

APPROVAL

This is to certify that Muhanguzi Amos (**REG NO: SEP23/BCS/3584U/F**) and Nakalembe Sarah (**REG NO: SEP23/BCS/3630U/H**) have submitted their proposal to the School of Computing and Engineering for examination under my guidance and supervision, in partial fulfillment of the requirements for the award of a Bachelor's Degree in Computer Science of Universal Technology and Management University.

Signature:

Date:

ALEX MUKAMA

School of Computing and Engineering

Universal Technology and Management University

TABLE OF CONTENTS

Contents

DECLARATION	iii
APPROVAL	iv
LIST OF ABBREVIATIONS AND ACRONYMS	vii
1.0 CHAPTER ONE: INTRODUCTION	1
1.1 Introduction	1
1.2 Background to the Study	1
1.3 Problem Statement	2
1.4 Research Objectives	3
1.4.1 General Objective	3
1.4.2 Specific Objectives	3
1.4.3 Research Questions	3
1.5 Research Scope	4
1.5.1 Geographical Scope	4
1.5.2 Subject Scope	4
1.5.3 Technical Scope	4
1.5.4 Time Scope	5
1.6 Research Significance	5
1.6.1 To Society	5
1.6.2 To Science and Technology	5
1.6.3 To Academia	5
1.6.4 To Industry and Government	5
1.7 Justification	6
1.8 Operational Definition of Terms	6
CHAPTER TWO: LITERATURE REVIEW	8
2.1 Introduction	8
2.2 Theoretical Review	8
i. Systems Theory	8
ii. Biometric Identity Management Theory	8
iii. Information Systems Success Model (DeLone & McLean, 1992)	9
iv. Predictive Analytics Theory	9
2.3 Conceptual Review	9
2.3.1 Biometric Systems in Institutional Contexts	9
2.3.2 Fingerprint Template Storage and Database Design	10
2.3.3 Role-Based Access Control and Receptionist Login	10
2.3.4 Artificial Intelligence in Prison Management	11

2.3.5 Web-Based Institutional Information Systems.....	11
2.4 Empirical Review.....	11
2.4.1 Current Administrative Context of Uganda Prisons Service (UPS)	11
2.4.2 Biometric Identification in Global Correctional Facilities.	12
2.4.3 Existing Commercial Prison Management Software	12
2.4.4 Analogous Biometric Institutional Deployments in Uganda.	13
2.4.5 Visitor Management and AI Security Analytics	13
2.5 Comparative Analysis.....	13
2.6 Synthesis of Literature.....	14
CHAPTER THREE: METHODOLOGY.....	16
3.0 Introduction	16
3.1 Requirement Gathering and Research Design.....	16
3.1.1 Research Design.....	16
3.1.2 Study Population and Sampling	16
3.1.3 Data Collection Methods	16
3.1.4 Data Analysis.....	17
3.2 System Design.....	17
3.2.1 System Requirements Summary.....	17
3.2.2 System Architecture	18
3.2.3 Biometric Workflow Design	18
3.2.4 Component Design.....	19
3.2.5 Database Schema Overview.....	20
3.2.6 Technology Stack Summary	20
3.2.7 Prototyping	21
3.3 System Development	21
3.3.1 Biometric Module Implementation.....	21
3.3.2 Backend and AI Implementation.....	22
3.3.3 Frontend Implementation	23
3.3.4 Development Practices.....	23
3.4 System Testing and Validation.....	24
3.4.1 Biometric Accuracy Testing	24
3.4.2 Functional Testing	24
3.4.3 Performance Testing	24
3.4.4 Security Testing.....	24
3.4.5 Usability Testing.....	25
3.5 Conclusion	25
REFERENCES	26

LIST OF ABBREVIATIONS AND ACRONYMS

Acronym	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BLOB	Binary Large Object
CSS	Cascading Style Sheets
CSRF	Cross-Site Request Forgery
ER	Entity Relationship
FAQ	Frequently Asked Questions
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JS	JavaScript
JWT	JSON Web Token
ML	Machine Learning
NIST	National Institute of Standards and Technology
ORM	Object-Relational Mapping
PMS	Prison Management System
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
SDK	Software Development Kit
SQL	Structured Query Language
UML	Unified Modeling Language
UI/UX	User Interface / User Experience
UTAMU	Uganda Technology and Management University
XSS	Cross-Site Scripting

1.0 CHAPTER ONE: INTRODUCTION

1.1 Introduction

This study focuses on the design and development of an AI-enhanced web-based Prison Management System (PMS) for managing inmate and visitor records at Luzira Prison, Uganda. A defining feature of the proposed system is the integration of biometric fingerprint technology at the point of inmate intake, enabling the system to automatically identify whether an individual being admitted has ever been held at the facility before. This capability allows the receptionist to instantly retrieve an existing inmate record upon a fingerprint match, pre-populate the intake form with historical data, and update it for the new admission significantly reducing data entry time and eliminating duplicate records.

Beyond biometrics, the system integrates Artificial Intelligence capabilities including prediction of inmate behavioural risk indicators, population Forecasting: Predicting future facility overcrowding by analysing sentencing trends, parole eligibility dates, and intake rates, Risk Profiling: By cross-referencing visitor data with inmate records, AI can flag visitors who have a high statistical probability of attempting to smuggle contrabands.

This chapter presents the background to the study, problem statement, research objectives, scope, significance, justification, and operational definitions.

1.2 Background to the Study

Prisons occupy a critical position within criminal justice systems globally, being responsible for the secure custody, welfare monitoring, and rehabilitation of inmates. Accurate and timely management of inmate and visitor records is foundational to fulfilling these responsibilities, underpinning security protocols, medical care, legal compliance, and administrative decision-making [1].

Globally, correctional facilities have increasingly adopted integrated digital systems combining biometric identification, centralized record management, and data analytics. Research documents that systems using biometric enrolment at inmate intake particularly fingerprint-based identification have substantially reduced duplicate record creation, accelerated inmate processing times, and improved the accuracy of recidivism

tracking in facilities across North America and Europe [2]. These systems provide the critical capability of instantly recognizing a returning inmate upon re-admission, automatically surfacing their prior record without manual search.

Across Sub-Saharan Africa, adoption of such integrated systems remains limited. Most correctional facilities in the region continue to rely on paper-based or minimally digitized records with no biometric capability, resulting in persistent challenges including misplaced files, duplicate records for repeat offenders, delayed retrieval, and inadequate data security [3]. The inability to quickly establish whether an admitted individual has a prior custody history at the same facility creates security risks and administrative inefficiencies that compound over time as inmate populations grow.

In Uganda, Luzira Prison stands as the largest correctional facility in the country and processes a high daily volume of inmate admissions, releases, transfers, and visitor interactions. Current approaches to managing this data reportedly combine paper records with disconnected basic digital files, lacking both integration and any form of automated identity verification. This means that when a formerly incarcerated individual is re-admitted, staff must manually search records a process prone to failure, particularly for individuals using different names or documentation.

1.3 Problem Statement

The management of inmate and visitor records at Luzira Prison currently depends on non-integrated, semi-digital methods that create significant operational and security deficiencies. The absence of any biometric identification capability at the point of inmate intake means that when an individual with a prior custody history is re-admitted, staff cannot automatically identify them or retrieve their existing record, resulting in duplicate files, lost historical data, and an inability to accurately track recidivism.

Additionally, the current system lacks role-based access controls, meaning that record retrieval and modification are inadequately secured against unauthorized access or tampering. The increasing volume of inmate admissions and visitor traffic at Luzira further strains administrative capacity, while the complete absence of intelligent data analysis tools prevents administrators from prediction of inmate behavioural risk indicators, population Forecasting: Predicting future facility overcrowding by analysing

sentencing trends, parole eligibility dates, and intake rates, Risk Profiling: By cross-referencing visitor data with inmate records, AI can flag visitors who have a high statistical probability of attempting to smuggle contrabands.

1.4 Research Objectives

1.4.1 General Objective

To design and develop an AI-enhanced web-based Prison Management System with biometric inmate and visitor identification for efficient management and intelligent analysis of inmate and visitor records at Luzira Prison, Uganda.

1.4.2 Specific Objectives

- i. To study and analyse the current methods used in managing inmate and visitor records at Luzira Prison and gather system requirements, including requirements for biometric identification at inmate intake.
- ii. To design a web-based Prison Management System architecture, biometric integration model, and database schema.
- iii. To develop and implement the web-based system integrated with biometric fingerprint functionality and AI modules for intelligent data analysis.
- iv. To test and evaluate the performance, biometric accuracy, security, and usability of the developed system.

1.4.3 Research Questions

- v. What are the current methods used in managing inmate and visitor records at Luzira Prison, and what biometric and digital capabilities are required to address their limitations?
- vi. How can a secure, biometric-capable web-based Prison Management System architecture and database model be designed to meet the identified requirements?
- vii. How can fingerprint biometrics and AI functionalities be integrated into the system to enable automated inmate identification, returning inmate record retrieval, and intelligent data analysis?

- viii. How can the developed system be tested and validated for biometric accuracy, performance, security, and usability?

1.5 Research Scope

1.5.1 Geographical Scope

This study is geographically limited to Luzira Prison, located in Kampala, Uganda, which serves as the primary case study. Luzira Prison is selected owing to its status as Uganda's largest correctional facility and the representativeness of its administrative challenges for the broader prison sector in the country.

1.5.2 Subject Scope

The study covers the management of inmate records (personal details, biometric enrolment, admission and release data, transfer history, and medical records) and visitor records (visitor identification, and visit logs). The biometric component is scoped to fingerprint-based identification for new inmate enrolment and returning inmate recognition at re-admission and visitors to. prediction of inmate behavioural risk indicators, population Forecasting: Predicting future facility overcrowding by analysing sentencing trends, parole eligibility dates, and intake rates, Risk Profiling: By cross-referencing visitor data with inmate records, AI can flag visitors who have a high statistical probability of attempting to smuggle contrabands.

1.5.3 Technical Scope

The system will be developed using HTML, CSS, Bootstrap, and JavaScript for the frontend; Python with the Flask framework for backend logic; and SQLite as the relational database for the prototype phase. Biometric integration will use a USB fingerprint scanner (compatible with DigitalPersona, ZKTeco, or equivalent) with the corresponding vendor Python SDK or the pyfingerprint open-source library for fingerprint capture, template extraction, and 1:N matching. Fingerprint templates will be stored as binary BLOB fields in the database. The AI module will use scikit-learn (Isolation Forest) for anomaly detection and pandas for report generation. Role-based access control (RBAC) will be implemented to manage user permissions for different staff categories such as (Receptionist, and System Administrator roles.)

1.5.4 Time Scope

The project will be completed within approximately three months, covering requirements gathering, system design, development and integration (including biometric module), testing and validation, and documentation.

1.6 Research Significance

1.6.1 To Society

The system will directly enhance Luzira working prison environment by improving the accuracy and speed of inmate identification at Luzira Prison. Biometric recognition of returning inmates will reduce the risk of false identity claims, ensure prior criminal histories are correctly associated with the right individual, and support more informed decisions on inmate classification and security placement.

1.6.2 To Science and Technology

This research contributes to the body of knowledge on biometric system integration in resource-constrained institutional settings in East Africa. It demonstrates a practical approach to combining fingerprint biometrics, web-based record management, and AI-driven analysis within a single cohesive system architecture.

1.6.3 To Academia

The project provides a documented, replicable model for biometric-enhanced institutional management systems applicable to correctional facilities and comparable public institutions in Uganda. It adds to academic literature on e-government, biometric identity management, and AI applications in criminal justice administration, supporting future research and innovation in these fields.

1.6.4 To Industry and Government

The project provides the Uganda Prisons Service with a concrete prototype and implementation roadmap for modernizing prison record management with biometric identification. The system architecture and methodology are designed with scalability in mind, making the solution extensible to other correctional facilities across Uganda and contributing to the broader digital transformation of public institutions.

1.7 Justification

The justification for this project is grounded in documented operational deficiencies at Luzira Prison and evidenced global trends toward AI-assisted correctional management. Research by [2] on prison visitation behaviour highlights the significance of structured visitor data analysis in understanding inmate conduct and institutional security, underscoring the value of an intelligent visitor record management module. [3] documents persistent file management challenges in prison administration across the East African context, directly corresponding to the operational problems identified at Luzira.

The Uganda Prisons Service's mandate to ensure secure custody and facilitate rehabilitation of inmates cannot be effectively fulfilled without reliable, accurate, and timely access to inmate and visitor data. The existing fragmented systems create preventable risks including unauthorized access, record tampering, and delayed responses to security incidents.

Furthermore, the availability of robust open-source web development frameworks (Flask, SQLite) and AI libraries (scikit-learn, pandas) biometric identification makes it technically and financially feasible to build a high-quality system within the resource constraints of an undergraduate final year project. Focusing on Luzira Prison as a case study provides a concrete, high-impact application context that can serve as a model for nationwide adoption by the Uganda Prisons Service.

1.8 Operational Definition of Terms

- ix. Prison Management System (PMS): A web-based software platform designed to centralize and automate the management of inmate and visitor records within a correctional facility, incorporating biometric identification and AI-driven analysis.
- x. Biometric Enrolment: The process of capturing a new inmate and visitor fingerprint using a fingerprint scanner and storing the resulting digital template in the system database, associating it permanently with visitor's inmate's record.
- xi. Biometric Matching (1: N): A fingerprint identification process in which a newly captured fingerprint is compared against all previously enrolled templates in the database to determine whether the individual has a prior custody history.

- xii. **Returning Inmate:** An individual being re-admitted to Luzira Prison who was previously enrolled in the system; recognized automatically through fingerprint matching, with their existing record retrieved for review and updating.
- xiii. **Fingerprint Template:** A mathematical representation extracted from a scanned fingerprint image, stored as a binary BLOB in the database, used for subsequent identity matching without storing the raw fingerprint image.
- xiv. **Receptionist / Intake Officer:** An authorized prison staff member responsible for processing new inmate admissions using the system, including triggering the biometric scan, reviewing retrieved records for returning inmates, and completing intake form data entry.
- xv. **Inmate Record:** A structured digital entry capturing personal, legal, medical, biometric, and behavioural information pertaining to an individual held in custody at Luzira Prison.
- xvi. **Visitor Record:** A structured digital log capturing identity verification, and interaction data for individuals granted access to visit inmates.
- xvii. **AI Module:** The Artificial Intelligence component responsible for prediction of inmate behavioural risk indicators, population Forecasting: Predicting future facility overcrowding by analysing sentencing trends, parole eligibility dates, and intake rates, Risk Profiling: By cross-referencing visitor data with inmate records, AI can flag visitors who have a high statistical probability of attempting to smuggle contrabands.
- xviii. **Role-Based Access Control (RBAC):** A security mechanism restricting system access and functionality to authorized users based on defined institutional roles such as Receptionist, Records Officer, and System Administrator.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature relevant to the design of an AI-enhanced, web-based Prison Management System (PMS) with integrated biometric identification. The review covers theoretical foundations underpinning biometric identity management and institutional information systems; conceptual frameworks for biometric integration, role-based access control, and AI-driven analytics; and empirical evidence from global correctional deployments alongside analogous institutional biometric frameworks within Uganda. The chapter concludes with a comparative analysis and a synthesis identifying the critical operational and technological gaps that the proposed system addresses.

2.2 Theoretical Review

i. Systems Theory

Systems theory holds that an institution achieves optimal efficiency when its component processes are integrated and mutually reinforcing rather than operating in isolation. Applied to prison management, this theory underpins the proposed design in which biometric identification, inmate records, visitor management, AI analytics, and administrative reporting function as tightly integrated modules within a single platform. This holistic approach eliminates the data silos, human errors, and manual handoffs that characterize the current fragmented environment at Luzira Prison.

ii. Biometric Identity Management Theory

Biometric identity management theory posits that physiological or behavioural characteristics that are unique, permanent, and measurable provide a more reliable basis for identity verification than knowledge-based (passwords) or possession-based (ID cards) tokens [4]. Fingerprints satisfy these criteria and have been established as the most widely deployed biometric modality in institutional settings globally due to their high distinctiveness, low-cost capture hardware, and mature matching algorithms. This theoretical framework directly justifies the selection of fingerprint biometrics for identity management within a correctional environment.

iii. Information Systems Success Model (DeLone & McLean, 1992)

The DeLone and McLean Information Systems Success Model evaluates system effectiveness across six dimensions: system quality, information quality, service quality, use, user satisfaction, and net benefits. This framework guides the design of the proposed PMS by emphasizing that the biometric and AI capabilities must not only function correctly but must deliver accurate, real-time data through an interface that prison staff find highly usable, trustworthy, and efficient. The model directly informs both the usability testing protocol and the validation criteria for the receptionist and administrative interfaces [5].

iv. Predictive Analytics Theory

Predictive analytics theory holds that historical institutional data contains embedded, non-obvious patterns that can be leveraged to forecast future behaviours or events. Integrated with criminological frameworks tracking recidivism (re-offending patterns), this theory justifies using historical inmate admission frequencies, cross-matched biographical details, and visitor interaction data to flag high-risk trends. As supported empirically by [2], automated predictive patterns aid institutional security by shifting management from a purely reactive posture to a proactive, data-informed strategy.

2.3 Conceptual Review

2.3.1 Biometric Systems in Institutional Contexts

Biometric systems in institutional settings typically follow a two-phase architecture: an enrolment phase where the biometric trait is captured and a template is digitally stored, and an identification or verification phase where a newly captured trait is compared against stored templates [6].

In a **1:N (one-to-many) identification scenario** (as required for returning inmate detection), the system compares a probe fingerprint against all enrolled templates in the database and returns the closest match above a defined confidence threshold. If no match is found, it triggers the new inmate enrolment workflow. For prison applications, 1:N identification is the operationally mandatory mode; returning inmates may attempt to present a false identity or alias, and the system must independently determine prior

custody records without relying on user-provided claims. This is a materially more demanding requirement than simple 1:1 verification (confirming that a claimed identity matches a single stored template).

The pyfingerprint Python library provides an open-source interface to ZKTeco-compatible fingerprint sensors, enabling template extraction, storage, and matching within a Python/Flask application. For higher-specification deployments, vendor SDKs from DigitalPersona (HID Global) provide advanced matching algorithm options available with Python bindings.

2.3.2 Fingerprint Template Storage and Database Design

Fingerprint templates are typically processed into minutiae feature vectors and stored as binary data in a BLOB (Binary Large Object) column within a relational database. SQLite natively supports BLOB storage, making it highly suitable for the prototype and testing phases[7].

Each inmate record in the proposed system carries a fingerprint_template BLOB field and a biometric_enrolled Boolean flag. During re-admission, the backend queries the enrolled templates, runs the 1:N match algorithm via the sensor interface, and on a positive match returns the matched inmate's unique Record ID. This automatically triggers the pre-population of historical intake forms, allowing the receptionist to verify, update, and commit the new admission without creating duplicate records.

2.3.3 Role-Based Access Control and Receptionist Login

Role-Based Access Control (RBAC) is the industry-standard security model for multi-user institutional systems (NIST SP 800-53) [8]. In the proposed PMS, five clear operational roles are defined:

- **System Administrator:** Full system access, configuration, and audit log inspection.
- **Records Officer:** Comprehensive inmate file and sentence record management.
- **Receptionist / Intake Officer:** Biometric capture, inmate intake processing, and visitor registration.

- **Warden:** Read-only access to all modules, system-wide overrides, and the AI analytics dashboard.

Each active session is authenticated using bcrypt-hashed credentials and managed via Flask-Login with JSON Web Tokens (JWT) carrying role-specific claims to gate access to protected application routes.

2.3.4 Artificial Intelligence in Prison Management

The application of machine learning to institutional behavioral data has demonstrated significant analytical value in correctional contexts. **Isolation Forest**, an unsupervised anomaly detection algorithm implemented via scikit-learn, is uniquely suited for identifying unusual visitor frequencies, irregular timing patterns, or suspicious visitor-to-inmate associations in high-dimensional institutional data without requiring pre-labeled training examples.

[2] documents strong empirical associations between structured visitation patterns and institutional security outcomes, providing a clear foundation for the visitor anomaly module. Furthermore, automated administrative report generation using pandas data aggregation minimizes the systemic administrative burdens and data-entry backlogs common in public registries.

2.3.5 Web-Based Institutional Information Systems

Flask-based web frameworks deliver highly reliable, lightweight, and secure institutional record management within developing infrastructure contexts [9]. The classic three-tier architecture HTML, CSS, Bootstrap, and JavaScript frontend, Flask backend controller, and an SQLite database layer offers a clear, low-overhead development pathway while remaining fully scalable to enterprise database engines.

2.4 Empirical Review

2.4.1 Current Administrative Context of Uganda Prisons Service (UPS)

While the Uganda Prisons Service (UPS) has established progressive institutional mandates under the **Prisons Act of 2006**, daily administration and custody tracking in major facilities like Luzira Prison still heavily rely on fragmented, offline desktop logging. This manual reliance creates acute operational vulnerabilities:

- **Alias Exploitation and Duplicate Profiles:** Inmates serving sentences under false names or hiding prior criminal histories cannot be reliably unmasked without centralized biometric duplication checks.
- **Siloed Data Architecture:** Inmate biographical profiles, medical records, and visitor logs exist in separate physical files, preventing real-time data aggregation and rapid administrative reporting[10].

This localized reality highlights that an integrated, web-based, biometric platform is a fundamental security and compliance necessity rather than a simple luxury.

2.4.2 Biometric Identification in Global Correctional Facilities.

Fingerprint-based inmate identification platforms have been extensively vetted across international correctional frameworks. In the United States, state departments of corrections (such as California and Texas) have long integrated fingerprint biometrics directly into Automated Fingerprint Identification Systems (AFIS) linked to their core management platforms. These deployments demonstrate both the operational stability and the absolute technical feasibility of 1:N fingerprint matching at scale, solving the core challenge of identity confirmation at points of intake, court transfer, and final release.

2.4.3 Existing Commercial Prison Management Software

Commercial enterprise suites, such as Tyler Technologies' Corrections software or OpenElms, offer deeply integrated inmate logging and biometric modules[11]. However, these proprietary systems are strictly designed for wealthy Western correctional systems. They are cost-prohibitive, demand high network infrastructure dependencies, and fail to align with the unique local administrative workflows of African correctional institutions. Currently, no purpose-built, open-source, or locally developed biometric-capable prison management system tailored to the budget parameters of the Ugandan context is documented in literature.

2.4.4 Analogous Biometric Institutional Deployments in Uganda.

Uganda's extensive deployment of biometric systems in other public sectors such as the National Identification and Registration Authority (NIRA) voter and ID registration, the UNHCR refugee registration framework, and mandatory mobile money SIM card verifications confirms that biometric hardware and enrolment workflows are highly viable in the local technological landscape. These deployments demonstrate that the local workforce possesses an established institutional familiarity with fingerprint capture devices, proving that deploying a biometric PMS at Luzira is highly realistic.

2.4.5 Visitor Management and AI Security Analytics

Vetting visitor data provides critical insights into institutional security. [2] compiled empirical evidence from European correctional facilities showing that structured analysis of visitation data; including shifts in frequency, unusual visit timing, and complex visitor-inmate networks; is meaningfully predictive of internal security and inmate conduct. Furthermore,[12] documented direct links between irregular visitation patterns and elevated risks of contraband introduction. These findings justify the inclusion of an AI-driven visitor anomaly detection module to generate risk indicators for prison staff.

2.5 Comparative Analysis

To clearly contextualize the value of the proposed system, the table below maps the structural features of the proposed AI-enhanced Prison Management System against existing manual and digital record-keeping alternatives:

Feature	Luzira prisons inmate counting system	Luzira Ward Allocation system (Basic Digital)	General Luzira PMS	Existing Luzira PMS + Biometrics (Inmates & visitors)	Proposed AI-PMS for Luzira.
Biometric Enrolment & Matching	No	No	No	No	Yes
Returning Inmate Auto-Retrieval	No	No	No	Partial	Yes
AI Anomaly Detection	No	No	No	No	Yes

Feature	Luzira prisons inmate counting system	Luzira Ward Allocation system (Basic Digital)	General Luzira PMS	Existing Luzira PMS + Biometrics (Inmates & visitors)	Proposed AI-PMS for Luzira.
Visitor Pattern Analysis	No	No	No	No	Yes
Automated Reporting	No	Partial	Partial	Partial	Yes
Centralized Records	No	No	partial	Yes	Yes
Web-Based Access	No	No	No	No	Yes
Receptionist Login (RBAC)	No	No	Partial	Partial	Yes
Inmate Re-Admission Workflow	No	No	No	Partial	Yes
Audit Trail	No	No	Partial	Partial	Yes

2.6 Synthesis of Literature

The reviewed literature consistently establishes that non-integrated record management systems in correctional environments create severe operational and security deficiencies, including profile duplication, undetected returning inmates, vulnerability to identity fraud, and a total lack of predictive security capabilities. Global evidence confirms that biometric-integrated management platforms optimize identity accuracy and administrative workflows, while modern AI algorithms provide highly valid operational insights from visitor patterns.

The definitive **knowledge and technological gap** lie in the complete absence of a cost-effective, purpose-built, biometric-capable, and AI-enhanced web system designed specifically around the operational realities, infrastructure constraints, and legislative framework (Prisons Act, 2006) of Ugandan facilities like Luzira Prison. Proprietary Western platforms are blocked by high financial barriers and rigid infrastructure requirements, while local biometric frameworks (like NIRA) are completely decoupled from correctional management.

The proposed system directly fills this critical gap. By combining open-source 1:N fingerprint biometric matching, a secure Python/Flask web controller protected by role-based access control, localized PDF reporting engines, and an unsupervised machine learning module for visitor anomaly detection, it delivers an actionable, accessible, high-security prototype tailored precisely for the Uganda Prisons Service environment.

CHAPTER THREE: METHODOLOGY

3.0 Introduction

This chapter presents the structured approach used to gather requirements, design, develop, test, and validate the AI-enhanced Prison Management System with integrated biometric identification. The methodology is organized around the four specific objectives of the study and follows the Software Development Life Cycle (SDLC) adapted to the academic project context.

3.1 Requirement Gathering and Research Design

3.1.1 Research Design

A mixed-methods research design is adopted, combining qualitative and quantitative data collection approaches to comprehensively characterize the current state of record management at Luzira Prison and establish functional, non-functional, and biometric system requirements.

3.1.2 Study Population and Sampling

The study population comprises administrative and security staff at Luzira Prison directly involved in inmate intake, record management, and visitor processing including reception officers, records officers, wardens. Purposive sampling will be used to select 10 to 20 key informants with direct experience of these processes. Snowball sampling will supplement recruitment where direct access is constrained.

3.1.3 Data Collection Methods

- Semi-structured interviews with intake officers and records staff to understand current workflows, identify pain points in inmate identification, and identify functional requirements for biometric integration.
- Observation of inmate intake and visitor check-in procedures to capture process realities not fully conveyed through interviews.
- Document review of existing record-keeping templates and forms to inform database field design.

- Structured questionnaires administered to a broader staff sample to quantify the frequency of record-keeping errors, duplicate record incidents, and perceived system requirements.

3.1.4 Data Analysis

Qualitative data will be analysed using thematic analysis to identify system requirement themes and workflow gaps. Quantitative data will be analysed using descriptive statistics to characterize the scale of identified challenges. Requirement findings will be directly mapped to system modules and biometric workflow specifications.

3.2 System Design

3.2.1 System Requirements Summary

Based on the requirements analysis, the system will support the following core capabilities:

- Receptionist login through a secure, role-authenticated web interface using bcrypt-hashed credentials and JWT session management.
- New inmate enrolment: capture of biographic and legal data through an intake form, followed by fingerprint scanning and template storage in the database.
- Returning inmate identification: upon fingerprint scan at re-admission, the system runs a 1:N match against all enrolled templates; a positive match retrieves the existing inmate record and pre-populates an editable intake form; a failed match opens the new inmate registration form.
- Editing and saving the pre-populated returning inmate form to update the record for the new admission episode.
- Visitor registration, visit logging, and identity verification.
- AI-driven inmate behavioural risk indicators, population Forecasting, risk Profiling by cross-referencing visitor data with inmate records, AI can flag visitors who have a high statistical probability of attempting to smuggle contrabands.
- Centralized administrator dashboard with real-time population and visitor statistics and AI alert display.
- Full audit trail logging all record creation, modification, and access events.

3.2.2 System Architecture

The system follows a three-tier client-server architecture:

- **Presentation Layer:** HTML, CSS, Bootstrap, and JavaScript-based web interface delivering role-specific dashboards and intake forms. The biometric capture interface will integrate with the fingerprint scanner via a lightweight browser-to-local-agent bridge (a small Python service running locally at the intake terminal that communicates with the scanner SDK and passes template data to the Flask backend via a secure local API call).
- **Application Layer:** Python Flask framework handling authentication, routing, biometric matching logic, AI module invocation, and database ORM operations.
- **Data Layer:** SQLite relational database storing all inmate records (including biometric templates as BLOB fields), visitor records, user accounts, audit logs, and AI analysis results.

3.2.3 Biometric Workflow Design

The biometric intake workflow is the central innovation of the proposed system and is designed as follows:

- **Step 1 — Receptionist Login:** The intake officer authenticates through the web login page using their assigned credentials. RBAC enforces that only the Receptionist role can access modules that defines his/her role in the system.
- **Step 2 — Fingerprint Scan Trigger:** The receptionist/intake officer initiates a new intake event from the dashboard. The system prompts the officer to place the inmate's finger on the connected scanner. The local Python agent captures the fingerprint image and extracts a digital template using the scanner SDK (e.g., pyfingerprint or ZKTeco SDK).
- **Step 3 — 1: N Database Matching:** The extracted template is transmitted to the Flask backend, which queries the Inmate table and executes a 1: N comparison of the probe template against all stored fingerprint_template BLOB records. A

configurable match confidence threshold (e.g., 75%) determines a positive identification.

- Step 4a — Returning Inmate Path: If a match is found, the system retrieves the matched inmate's full record, pre-populates the intake form with all existing data (personal details, prior admission history, medical flags, disciplinary log), and presents it to the receptionist for review. The officer can edit any fields (e.g., updated next-of-kin, new legal charge) and saves the record, which creates a new admission episode entry linked to the existing inmate ID. A status update marks the inmate as currently admitted.
- Step 5a — Returning Inmate Confirmation: The system logs the re-admission event in the audit trail, timestamps the new episode, and updates the inmate's status dashboard entry.
- Step 4b — New Inmate Path: If no match is found, the system opens a blank new inmate registration form. The receptionist completes all required fields. Upon saving, the system stores the new inmate record and associates the captured fingerprint template with it in the database.
- Step 5b — New Enrolment Confirmation: The system assigns a unique Inmate ID, logs the enrolment event in the audit trail, and adds the new inmate to the active population dashboard.

3.2.4 Component Design

- Inmate Records Module: Manages complete inmate profiles including biographic data, legal status (charge, sentence, expected release date), admission history (all episodes), transfer records, medical records, and disciplinary logs. Database: Inmate table with fields inmate_id (PK), full_name, date_of_birth, gender, nationality, crime_category, first_admission_date, current_admission_date, expected_release_date, cell_block, medical_notes, status (active/released/transferred), fingerprint_template (BLOB), biometric_enrolled (BOOLEAN) among others.
- Visitor Management Module: Handles visitor registration with identity verification, visit scheduling, and gate check-in logging. Includes AI anomaly flag display on

visitor profiles. Database: Visitor and Visit_Log tables with visitor_id, inmate_id, visit_date, officer_id, and anomaly_flag fields among others.

xix. AI Analysis Module: Isolation Forest anomaly detection on visitor frequency time-series data; behavioral risk scoring aggregating disciplinary, population forecasting, flagging visitors who have a high statistical probability of attempting to smuggle contrabands. Database: AI_Log table with analysis_id, analysis_type, target_id, result, confidence_score, and timestamp among others.

- Administrative Dashboard: Real-time inmate population counts by status and cell block, visitor log summary, pending AI alerts, and report download links. Role-differentiated views for each staff category.
- User Management Module: RBAC with five defined roles. User accounts with bcrypt-hashed passwords. Flask-Login session management with JWT tokens. Login audit logging for all authentication events.
- Audit Trail Module: Immutable log of all system events including logins, record creation, modification, deletion, biometric scans, and AI analysis runs, with timestamps, user IDs, and IP addresses.

3.2.5 Database Schema Overview

The Entity-Relationship (ER) diagram will be created in Lucidchart. Core entities and key relationships: Inmate (one) — has many — Admission_Episode; Inmate (one) — has many — Visit_Log (via Visitor); User_Account (one) — has role — Role; User_Account (one) — generates many — Audit_Event; Inmate (one) — has one — Biometric_Template (embedded as BLOB in Inmate table); Inmate (one) — has many — Medical_Record; among others .

3.2.6 Technology Stack Summary

The complete technology stack for the proposed system is summarized in the table below:

Layer / Component	Technology / Tool	Purpose
Frontend	HTML5, CSS3, Bootstrap, JavaScript	Responsive web UI, forms, dashboards
Backend	Python 3, Flask Framework	Server logic, routing, authentication, API endpoints
Database	SQLite (prototype)	Inmate records, visitor logs, biometric BLOB storage, audit trail
Biometric Capture	Vendor SDK (e.g., ZKTeco / DigitalPersona) + pyfingerprint Python library	Fingerprint enrolment and 1: N matching at inmate intake and re-admission
AI / ML	scikit-learn, pandas, NumPy, Matplotlib	Anomaly detection (Isolation Forest), risk scoring, automated reports
Security	Flask-Login, JWT, bcrypt, Flask-WTF (CSRF), HTTPS	Session management, password hashing, access control, CSRF protection
Version Control	Git + GitHub	Collaborative development and code versioning
Deployment	Render.com (free tier)	Web hosting for prototype testing and validation
Design / Prototyping	Figma, Lucidchart	UI mock-ups and UML / ER diagrams

3.2.7 Prototyping

High-fidelity UI mock-ups will be developed in Figma covering the receptionist login screen, biometric intake workflow screens (scan prompt, returning inmate match result, new inmate registration form), inmate record view and edit form, visitor management interface, and administrator dashboard with AI alert panel. These mock-ups will be reviewed with selected prison staff before development begins.

3.3 System Development

3.3.1 Biometric Module Implementation

- **Hardware Setup:** A USB fingerprint scanner (ZKTeco ZK4500 or equivalent available in Uganda) connected to the intake terminal. The pyfingerprint Python library or vendor SDK installed on the local intake machine [13].
- **Local Biometric Agent:** A lightweight Python Flask microservice running on the intake terminal that interfaces directly with the scanner SDK. On request from the main web application, it captures a fingerprint, extracts the template, and returns it via a secure local HTTP call to the main Flask backend.

- **1: N Matching Algorithm:** The Flask backend retrieves all enrolled fingerprint_template BLOBs from the Inmate table, iterates through them using the SDK's comparison function (or a library such as python-fingerprint-recognition for template comparison), and returns the inmate_id of the record with the highest match score above the defined confidence threshold. If no record exceeds the threshold, the result is returned as 'no match'.
- **Enrolment Flow:** On new inmate registration, after the receptionist completes the intake form and triggers enrolment, the biometric agent captures three fingerprint samples, the SDK generates a consolidated template from the three captures for improved accuracy, and the template is stored as a BLOB in the new inmate's database record.
- **Re-Admission Flow:** The 1: N match result triggers conditional routing in the Flask backend: a match returns a JSON payload containing the matched inmate's full record for pre-population of the frontend intake form; no match triggers the empty new registration form. The receptionist edits the pre-populated form as needed and submits, which the backend processes as a new Admission_Episode record linked to the existing inmate_id.

3.3.2 Backend and AI Implementation

- **Inmate and Visitor CRUD:** Flask routes with SQLAlchemy ORM for all create, read, update, and delete operations. Input validation on all form fields. Parameterized queries via ORM to prevent SQL injection.
- **RBAC Enforcement:** Flask decorator functions on all protected routes check the authenticated user's role claim from the JWT and return HTTP 403 for unauthorized access attempts.
- **AI Anomaly Detection:** scikit-learn Isolation Forest model trained on visitor frequency time-series extracted from the Visit_Log table. Model runs nightly via APScheduler, flags visitor records with anomaly scores below the contamination threshold, and writes results to the AI_Log table. Dashboard alert panel queries AI_Log for unreviewed flags.

- Automated Reporting: pandas aggregation queries on Inmate and Visit_Log tables generate weekly population summaries and visitor frequency reports, rendered as Flask HTML templates and offered as PDF downloads via WeasyPrint.

3.3.3 Frontend Implementation

- Responsive HTML/CSS/Bootstrap interface with role-specific navigation menus showing only accessible modules.
- Biometric intake page with real-time scan status indicator (waiting, scanning, processing, match found / no match found) communicating with the local biometric agent via AJAX.
- Returning inmate form pre-population using JavaScript to populate form fields from the JSON payload returned by the backend on a positive biometric match, with all fields remaining editable before submission.
- AI alert notifications displayed on the administrator and warden dashboard as a dismissible alert panel.

3.3.4 Development Practices

- Agile Methodology: Scrum framework with one-week sprints. Sprint priorities: Sprint 1 (requirements and design), Sprint 2 (authentication and RBAC), Sprint 3 (biometric module and inmate records), Sprint 4 (visitor management), Sprint 5 (AI module), Sprint 6 (dashboard and reporting), Sprint 7 (testing and validation).
- Secure Coding: CSRF protection via Flask-WTF, input validation on all endpoints, bcrypt password hashing, HTTPS enforcement in deployment configuration, secure cookie attributes.
- Version Control: Git with GitHub for all code and documentation, with feature branching for module development.

3.4 System Testing and Validation

3.4.1 Biometric Accuracy Testing

The biometric module will be validated using a controlled test dataset of fingerprint samples enrolled during development. Testing will measure the False Accept Rate (FAR incorrectly matching different individuals), False Reject Rate (FRR — failing to match the same individual), and overall identification accuracy. The match confidence threshold will be calibrated to optimize the FAR/FRR trade-off appropriate for a security-sensitive prison context (prioritizing low FAR). The returning inmate retrieval workflow will be tested end-to-end with both positive match scenarios (same individual re-scanned) and negative match scenarios (new individuals) to confirm correct form routing in both cases.

3.4.2 Functional Testing

All system modules will be tested using Flask's test client with Python unit test. Specific test cases will cover: receptionist login with valid and invalid credentials; biometric scan triggering and result routing (match and no-match paths); new inmate registration and record persistence; returning inmate form pre-population accuracy; form editing and save success; visitor registration and visit logging; RBAC enforcement across all protected routes; audit trail entry creation for all logged events.

3.4.3 Performance Testing

Load testing with Locust will simulate concurrent multi-user access, including simultaneous biometric scan requests, to assess system responsiveness under realistic operational loads. The 1: N matching performance will be benchmarked against a database of 500 enrolled templates (representative of a medium-sized prison population) to confirm sub-second identification response times.

3.4.4 Security Testing

- Static code analysis using Bandit to identify insecure Python coding patterns.
- Penetration testing simulating SQL injection, XSS, and CSRF attacks on all input endpoints.
- RBAC validation testing unauthorized access attempts across all role-protected routes.
- Supervisor security audit against OWASP Top 10 principles{Citation}.

- Biometric template security review to confirm templates are stored as non-reversible binary data (not raw images) and that database access to the BLOB field is restricted to backend processes only.

3.4.5 Usability Testing

Selected prison reception and administrative staff will interact with the prototype system, completing defined tasks including a simulated new inmate intake, a returning inmate re-admission, a visitor check-in, and an administrative report retrieval. Structured usability questionnaires will capture task completion rates, error frequencies, and satisfaction scores. Feedback will be analysed and prioritized interface improvements will be implemented before final submission.

3.5 Conclusion

The proposed AI-enhanced web-based Prison Management System addresses a clear and documented operational gap at Luzira Prison by integrating three capabilities that are currently entirely absent: biometric fingerprint identification for automated recognition and re-admission of returning inmates, a secure role-based login architecture ensuring appropriate access control for all staff categories, and AI-driven analytical capabilities for visitor anomaly detection and automated reporting. Built on an accessible, open-source technology stack that is practically achievable within the scope of an undergraduate final year project, the system provides both a functional prototype for immediate evaluation and a scalable foundation for potential adoption across the Uganda Prisons Service.

REFERENCES

- [1] M. F. Aebi and E. Cocco, "Prisons and prisoners in Europe 2023: Key findings of the SPACE I report," 2024, Accessed: Jul. 15, 2026. [Online]. Available: <https://iris.unil.ch/bitstreams/3e414f64-b77e-46c8-8020-c71386d7cflc/download>
- [2] M. A. ALMAIAH, A. ALI, T. T. TIN, T. ALKHDOUR, A. LUTFI, and M. ALRAWAD, "Unlocking user privacy: a privacy-focused cryptocurrencies framework for concealing transactions using zero-knowledge proofs (ZKPs)," *J. Theor. Appl. Inf. Technol.*, vol. 102, no. 8, p. 2024, 2024.
- [3] M. L. Berghuis, M. Sentse, H. Palmen, and P. Nieuwbeerta, "Receiving visits in prison and aggressive and contraband misconduct among Dutch prisoners," *Eur. J. Criminol.*, vol. 20, no. 4, pp. 1369–1389, Jul. 2023, doi: 10.1177/14773708211041016.
- [4] M. L. Berghuis, J. C. Cochran, P. Nieuwbeerta, and H. Palmen, "Do Prison Visits Have a Spectrum? Visitation Quality and the Varying Impacts for Individuals' Behavior and Well-Being in Prison," *Crim. Justice Behav.*, vol. 52, no. 5, pp. 671–689, May 2025, doi: 10.1177/00938548241291004.
- [5] W. H. DeLone and E. R. McLean, "Information Systems Success: The Quest for the Dependent Variable," *Inf. Syst. Res.*, vol. 3, no. 1, pp. 60–95, Mar. 1992, doi: 10.1287/isre.3.1.60.
- [6] A. K. Jain, A. A. Ross, and K. Nandakumar, "Face Recognition," in *Introduction to Biometrics*, Boston, MA: Springer US, 2011, pp. 97–139. doi: 10.1007/978-0-387-77326-1_3.
- [7] "A geometric transformation to protect minutiae-based fingerprint templates - art. no. 65390E." Accessed: Jul. 15, 2026. [Online]. Available: https://www.researchgate.net/publication/228804475_A_geometric_transformation_to_protect_minutiae-based_fingerprint_templates_-_art_no_65390E
- [8] S. NIST, "800-53; Security and Privacy Controls for Information Systems and Organizations," *Natl. Inst. Stand. Technol. NIST Gaithersburg MD USA*, 2020.
- [9] "FinTech, Financial Inclusion, and Sustainable Development | Disruption." Accessed: Jul. 15, 2026. [Online]. Available: <https://www.taylorfrancis.com/books/mono/10.4324/9781032657981/fintech-financial-inclusion-sustainable-development-david-mhlanga>
- [10] E. Hailemariam Ambaye, "Prisoners File Management", Accessed: Jul. 15, 2026. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5631793
- [11] L. O. Limyoco Jr, "Enhancing efficiency and security: A study of automated visitors monitoring system in the Bureau of Jail Management and Penology," *Int. J. Sci. Technol. Eng. Math.*, vol. 5, no. 4, pp. 44–63.
- [12] V. C. Iwuoha and M. Doevenspeck, "Biometric coloniality: digital consensus and the biometric state in Africa," *Third World Q.*, vol. 46, no. 12, pp. 1413–1438, Aug. 2025, doi: 10.1080/01436597.2025.2539731.
- [13] B. Raschke, *bastianraschke/pyfingerprint*. (Jun. 04, 2026). Python. Accessed: Jul. 15, 2026. [Online]. Available: <https://github.com/bastianraschke/pyfingerprint>